Ideal ignition windows for p-^{11}B in a 0-D power balance: no window at $T_e = T_i$ and thresholds on T_e/T_i, with two reactivity fits

(interval-arithmetic enclosures of a model statement; not a statement about any device)

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AI-assistance disclosure and audit status. This preprint, its Python and Lean code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was independently audited by a different AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** The audit's nine corrections are applied in this text (Section 8). The auditor's own recomputation was interval arithmetic for the Nevins–Swain and D-T quantities and *float only* (Tier [X]) for everything on the Wang side and for the D-T minimisers of the gate model. No human audit and no statement lock exist; both are owed (Section 10).

Abstract

We study the ideal-ignition criterion $P_c \geq P_B$ (fusion power against bremsstrahlung, no transport loss) of the 0-D power-balance model of Wurzel and Hsu [1] for proton–boron-11, with two reactivity fits: Nevins–Swain [5], as transcribed from the Wurzel–Hsu public code (MIT licence, commit `4ae563bf`), and the updated S-factor of Wang et al. [2]. In the box $T_i \in [50, 500]$ keV, boron fraction $x \in [0.05, 0.5]$, interval-arithmetic enclosures (Arb balls, Tier [C], not replayed in Lean) give: **there is no ignition window at $T_e = T_i$**: the certified maximum of P_c/P_B is in $[0.4214, 0.4256]$ (Nevins–Swain) and $[0.4287, 0.4326]$ (Wang). A window requires T_e/T_i below a threshold θ^*, certified only for $\theta \leq 1$: Nevins–Swain $[0.4223, 0.4308]$ over the box and $[0.3753, 0.3828]$ at $x = 1/6$; Wang $[0.4381, 0.4470]$ and $[0.396, 0.402]$. These thresholds hold for *these fits and this bremsstrahlung only*: a uniform factor 1.1/1.2/1.32 on the reactivity moves the threshold to 0.477/0.524/0.579 (auditor, float). The "no window at $T_e = T_i$" statement is more robust: it survives any reactivity up to 2.349 times the Nevins–Swain fit (certified). As a positive control the D-T ideal-ignition temperature and the Table II/III minima of [1] are reproduced to the printed digits *for the model the authors' code ran*; the gate model was chosen after a float run, and the most literal text reading of the paper prints 3.0×10^{21} instead of 2.9×10^{21} for the triple-product minimum. The Bosch–Hale and Nevins–Swain coefficients **could not be obtained from the primary papers by the producer or by the auditor** and are **not verified**; this matters most for θ^*. Two small Lean facts ($20x(1-x) \leq (1+4x)^2$ and $1 + 2.2t + 3t^2 - 4.375t^{2.5} > 0$ on $[0, 1]$) are kernel-checked, independently recompiled and replayed by the auditor, and are Tier [A] once a statement-lock entry exists (none yet).

What this is NOT. Every statement concerns an *idealised* 0-*D power-balance model with the stated fitted formulas*. It is **not** a statement about any reactor, device, cost, climate or economics, **not** a claim that p-^{11}B fusion is possible or impossible, **not** a claim that $T_e < T_i$ can be maintained, and **not** a discovery beyond the model. Not in the model: synchrotron radiation, transport and conduction losses, any ion–electron coupling mechanism, α heating of electrons, ash and impurities, non-Maxwellian or kinetic effects, and the uncertainties of the fits.

Markers. [A]: kernel-checked Lean, axioms within {`propext`, `Classical.choice`, `Quot.sound`}, controls caught. [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked (here: Arb interval arithmetic). [X]: a float check or a recorded absence, supporting nothing by itself. Identifiers `PBI-·-nnnn` refer to `ledger/ledger.json`.

1 Model and question

The model (Tier [L]; `PBI-L-0001`). As in Wurzel and Hsu [1] (eqs. (6)–(8), (16), (22), (41), (D1)–(D5)): the fusion power density is $S_c = f_c\, n_1 n_2 \langle\sigma v\rangle E_F$ and the bremsstrahlung loss is $S_B = C_B\, n_e^2 \sqrt{T_\mathrm{e}}\, \gamma(Z_\mathrm{eff})$ with $C_B = 5.34 \times 10^{-37}$ W m^3 keV$^{-1/2}$ and the relativistic bracket

$$\gamma = Z_\mathrm{eff}\big(1 + 1.78\, t^{1.34}\big) + 2.12\, t\big(1 + 1.1\, t + t^2 - 1.25\, t^{2.5}\big), \qquad t = T_\mathrm{e}/m_e c^2,$$

(their eq. (D2), equal to eq. (14) of Ochs and Fisch [3], which includes the x^2 term). Ideal ignition means $P_\mathrm{c} \geq P_\mathrm{B}$, with ideal confinement and no conduction loss. For p-^{11}B: $f_c = 1$, $E = 8.68$–8.70 MeV, boron fraction $x = n_B/n_i$, $n_e = (1+4x)n_i$, $Z_\mathrm{eff} = (1+24x)/(1+4x)$. The ratio $R = P_\mathrm{c}/P_\mathrm{B}$ does not depend on density. The Wurzel–Hsu optimal mix is $x = 1/6$ ($Z_\mathrm{eff} = 3$; their eqs. (D3)–(D4), footnote 138). The temperature ratio $\theta = T_\mathrm{e}/T_\mathrm{i}$ is a *free parameter*; no mechanism that would keep $T_\mathrm{e} < T_\mathrm{i}$ is modelled.

The question (preregistered; `docs/proposals/NEXT_STEPS_PROPOSAL_4_2026-10-27.md` and the addendum of the same date, item E1). (a) Reproduce the D-T ideal-ignition numbers of [1] as a positive control, with a gate fixed before the certified run. (b) For p-^{11}B in the box $T_\mathrm{i} \in [50, 500]$ keV, $x \in [0.05, 0.5]$: certify the threshold θ^* above which $P_\mathrm{c} < P_\mathrm{B}$ everywhere. (c) Certify whether any window exists at $T_\mathrm{e} = T_\mathrm{i}$, under the older fit and under an updated fit. The preregistration asked for Tier [A] inequalities and for coefficients pinned from the primary papers; neither was achieved (Section 10, Deviations 1 and 3).

2 Fits and what could not be verified

Reactivity fits. (i) *Nevins–Swain [5]*: the high-temperature fit, valid 50–500 keV, with coefficients $P_1, \ldots, P_7$, E_G, $M_r c^2$ and a resonant term, as transcribed in the Wurzel–Hsu public code at commit `4ae563bf` (MIT licence; the commit is dated after the 2022 paper). (ii) *Wang et al. [2]*: an updated S-factor in three segments (Table I of that paper), made a reactivity by the Maxwellian integral $\langle\sigma v\rangle = (8/\pi)^{1/2}\mu^{-1/2}T^{-3/2}\int_0^\infty \sigma E\, e^{-E/T}\, dE$, with $\sigma := 0$ above 10 MeV (their stated fit range). The first Wang segment reprints the Nevins–Swain low-energy coefficients, which independently corroborates that part of the transcription. All coefficients are decimal strings in `data/fits.json` with per-block provenance.

Not verified against the primary papers (Tier [L], `PBI-L-0003`; audit correction C9). Neither the producer nor the independent auditor could obtain Bosch–Hale (1992) [4] or Nevins–Swain (2000) [5] (closed access or bot-challenged). The Bosch–Hale (D-T) and Nevins–Swain (p-^{11}B) coefficients are therefore *secondary*: they are the values in the authors' code. The only checks are indirect: the transcribed Nevins–Swain fit divided by the Maxwellian integral of the transcribed Nevins–Swain S-factor is 0.985–0.996 for $T_\mathrm{i} = 50$–200 keV, 0.976 and 0.965 at 250 and 300 keV, 0.940 at 400 and 0.917 at 500 keV (producer; reproduced by the auditor; Tier [X]). This excludes gross transcription errors but cannot detect an error of a few per cent, and the high-temperature drift is unexplained. The auditor's agreement with recalled Bosch–Hale values is memory, not a pinned source. **Because θ^* moves roughly in proportion to the reactivity, it inherits any error in the unverified digits directly**; the $T_\mathrm{e} = T_\mathrm{i}$ exclusion has a margin of about 2.35 and does not.

Other sources. Read in full: Wang et al. (2601.00241) and Ochs–Fisch (2310.18508), via alphaXiv. Read in part: Wurzel–Hsu (physics sections and Appendices C/D; data tables not read). Sikora–Weller [6] (J. Fusion Energy 35, 538) was read but gives no reactivity fit and is not used as one. Putvinski et al. (2019) was seen only through OCR summaries (bremsstrahlung eq. (7) is consistent with the two texts read). Tentori–Belloni (2023) could not be accessed. Third-party full texts, the Wurzel–Hsu code and the paywall pages are *not* in this bundle (Section 8, C7).

3 Method

Arb enclosures (Tier [C]; `PBI-C-0001`). `data/code/e1_model.py` implements the model with python-flint 0.6.0 Arb balls at 128 bits; every coefficient is a decimal string. For a cross-section fit the Maxwellian integral is a rigorous `acb.integral` on each analytic segment plus a bound on $(0, 1]$ keV; a reactivity enclosure

over a temperature interval $[a, b]$ follows from point integrals by $\langle\sigma v\rangle(a)(a/b)^{3/2} \le \langle\sigma v\rangle(T) \le \langle\sigma v\rangle(b)(b/a)^{3/2}$, valid because $\sigma \ge 0$ (certified). The p-^{11}B reduced mass is enclosed in $\mu c^2 \in [859500, 859650]$ keV. The ignition temperature is a bisection plus certified signs on the outer intervals; minima use an upper bound at the float minimiser and certified lower bounds elsewhere; $\max R$ over (T_i, x) boxes is a best-first branch and bound with outward rounding; θ^* is a bisection with a two-sided certificate (certified $\max R < 1$ at θ_hi, and a point witness $R > 1$ at θ_lo).

From the face θ_hi to all $\theta \in [\theta_\mathrm{hi}, 1]$. $P_\mathrm{B} \propto \sqrt{T_\mathrm{e}}\,\gamma(T_\mathrm{e}/m_e c^2)$ is increasing in T_e for $T_\mathrm{e} \le 500$ keV$< m_e c^2$: the Z_eff term is increasing, and the derivative of the electron–electron term $t(1 + 1.1t + t^2 - 1.25t^{2.5})$ is $1 + 2.2t + 3t^2 - 4.375t^{2.5} > 0$ on $[0, 1]$, certified twice, by Arb subdivision and by the Lean lemma of Section 7. The step "positive derivative implies monotone" is not formalised (Tier [C]). Hence a certified $\max R < 1$ at θ_hi gives $R < 1$ for every $\theta \in [\theta_\mathrm{hi}, 1]$. **We certify nothing for $\theta > 1$** (there T_e can leave the 500 keV box toward $m_e c^2$ where the fit and the argument stop; Deviation 4).

Independent float code. `e1_float_indep.py` is a separate float implementation (scipy `quad`, `brentq`, text constants); it lies inside every certified enclosure (Tier [X], `PBI-X-0001`).

Run history. Runs 1 and 2 of the D-T stage and two p-^{11}B stage runs were aborted (hang, too wide, a grid edge rounding below 50 keV); run 3 failed G4 only, for the constant-hull reason. The final hashed `e1_certify.py` was rerun end to end (`logs/certify.json`, `logs/certify_final_stdout.txt`) and reproduces every gate verdict and enclosure quoted here; earlier logs are kept as history. In *decide* mode an early-exit bug could return "undecided" but not a false certificate; Wang at $x = 1/6$ is certified by the explicit scan `e1_wang_mix.py`, because the stopping order of `theta_threshold` (rtol test before the decide test) is still unfixed in the hashed code, so that the hashed code is exactly what ran.

4 Positive control: D-T (Tier [C]; PBI-C-0002)

The gate (`e1_gate.py`) was written before the certified run but *after a first float run* (Deviation 6). **It passes for the model the authors' code ran**: the `integrated` path (Maxwellian average of the Bosch–Hale cross section over $E \in [0.5, 4700]$ keV, certified with Arb), the Putvinski relativistic bracket with $Z_\mathrm{eff} = 1$, $f_c = 0.2$, $E = 17.6$ MeV and C_B from the NRL value 1.69×10^{-32}. (Paper source: github.com/swurzel/lawson-criterion-paper at `4ae563b`; the paper's Fig. 4 caption says the D-T reactivities come from the integrated cross sections.)

quantity	W–H printed	certified, authors' code model	constant hull
ideal ignition T (eq. 8, Fig. 6)	4.3 keV	[4.2989514, 4.2989523]	[4.2608, 4.3468]
argmin $n_i T\tau_E$, $Q = \infty$ (Tab. III)	14 keV	[14.128, 14.442]	corners [14.127, 14.448]
min $n_i T\tau_E$	2.9×10^{21}	[2.94410, 2.94502]$\times 10^{21}$	[2.94400, 2.96351]$\times 10^{21}$
argmin $n_i\tau_E$ (Tab. II)	26 keV	[25.590, 26.212]	corners [25.590, 26.214]
min $n_i\tau_E$	1.6×10^{20}	[1.55227, 1.55279]$\times 10^{20}$	[1.55224, 1.56219]$\times 10^{20}$

The constant hull combines C_B and the keV-to-joule conversion from the Wurzel–Hsu text (5.34×10^{-37}; 1.6×10^{-16} J/keV), from the code (NRL, 6.24×10^{18} or 6.242×10^{18} eV/J, CODATA), and the Ochs–Fisch prefactor, together with $f_c \in [3.5/17.6, 0.2]$; the minima are evaluated at the corners of the hull, valid because the triple-product minimum is monotone in f_c and in the bremsstrahlung constant.

Other certified readings, relabelled per audit correction C1. The variants formerly called "text constants" use the text's C_B and keV-to-joule conversion but the *code's* $f_c = 0.2$; they are labelled accordingly.

variant	$T_{\rm ign}$ (keV)	min $n_iT\tau_E$ (10^{21})	prints
parameterized σv, non-rel., text C_B/J, code $f_c = 0.2$	[4.29959, 4.29959]	[2.94923, 2.95056], straddles the 2.95 edge	2.9 or 3.0
parameterized, relativistic, code constants	[4.32619, 4.32620]	[2.96275, 2.96409]	3.0
integrated, non-rel., text C_B/J, code $f_c = 0.2$	—	[2.93077, 2.93168]	2.9
integrated, relativistic, code constants (the gate model)	—	[2.94410, 2.94502]	2.9
pure text (all text constants incl. $f_c = 3.5/17.6$), parameterized, non-rel. (auditor, [X])	[4.3071225, 4.3071227]	≥ 2.95 everywhere, value 2.96846 at 14.2486 keV; so in $[2.95, 2.96846]$	**3.0**

Every variant gives 4.3 keV, 14 keV, 26 keV and 1.6×10^{20}; only the second significant digit of the Table III minimum depends on the reading. The last row was computed by the auditor with `mpmath.iv` (interval arithmetic for $T_{\rm ign}$ and for the lower bound 2.95×10^{21} and the value at 14.2486 keV; the minimiser and the LP minimum 1.567×10^{20} at 25.98 keV are float). It is *not* certified by the producer's pipeline, and the auditor's scratch code is not shipped, so it is filed Tier [X] here (`PBI-X-0001`).

Honest status of the gate. The reproduction "to the printed digits" holds for the model the authors' code ran, not for every reading of the text. The gate window for G2 is $[2.85, 2.95) \times 10^{21}$, fixed after the float run showed that another reading rounds to 3.0×10^{21} (Deviation 6, disclosed). The Table III minimum sits 0.17% below the rounding edge 2.95×10^{21}.

5 Results for p-^{11}B (Tier [C])

5.1 No ignition window at $T_{\rm e} = T_{\rm i}$ (`PBI-C-0003`)

fit, region	certified max $P_{\rm c}/P_{\rm B}$	argmax $(T_{\rm i}, x)$	m^*
Nevins–Swain, box	[0.42139, 0.42556]	(202 keV, 0.100)	[2.350, 2.373]
Wang, box	[0.42869, 0.43260]	(208.5 keV, 0.101)	[2.312, 2.333]
Nevins–Swain, $x = 1/6$	[0.39016, 0.39240]	(204.5 keV)	[2.548, 2.563]
Wang, $x = 1/6$	[0.39732, 0.40083]	(211.4 keV)	[2.495, 2.517]

So $P_{\rm c} < P_{\rm B}$ everywhere on the box under both fits: no ideal-ignition window at $T_{\rm e} = T_{\rm i}$ in this ideal 0-D model (no synchrotron, no transport, Maxwellian, no mechanism maintaining $T_{\rm e} < T_{\rm i}$), for $T_{\rm i} \in [50, 500]$ keV and $x \in [0.05, 0.5]$ (audit correction C5). The maximum is interior (about 202 keV, $x \approx 0.10$) and widening the box ($x \in [0.005, 0.95]$; for Wang also $T_{\rm i} \in [20, 1000]$ keV) leaves it unchanged (auditor, float). The multiplier m^* is a certified statement that needs no updated fit: R is linear in $\langle\sigma v\rangle$, so any reactivity with $\langle\sigma v\rangle(T) \leq 2.349\times$ (Nevins–Swain)(T) on $[50, 500]$ keV gives $P_{\rm c} < P_{\rm B}$ on the whole box at $T_{\rm e} = T_{\rm i}$.

Caveat on comparing the two fits. Wang is an *integrated* S-factor; Nevins–Swain is a *fitted* reactivity. The Nevins–Swain fit lies 2.4–3.5% below the Maxwellian integral of the Nevins–Swain S-factor at 250–300 keV, where θ^* is set, and 8% below at 500 keV, so part of the Wang/Nevins–Swain difference in θ^* is method mismatch, not new data. The certified ratio Wang/NS of the reactivity is 1.016, 1.000, 1.005, 1.016, 1.070, 1.154, 1.262 at $50, 100, 150, 200, 300, 400, 500$ keV (point enclosures of relative radius below 10^{-4}); at 500 keV the Wang fit includes the new 4.7 MeV resonance. The $T_{\rm e} = T_{\rm i}$ exclusion does not depend on this (margin about 2.3 either way). Wang et al. themselves state, with Rider's bremsstrahlung, that with $T_{\rm e} = T$ the bremsstrahlung exceeds the fusion power over the whole temperature range (Tier [L], `PBI-L-0002`); this agrees with, and is not an independent check of, the Putvinski-bremsstrahlung result here.

5.2 Thresholds on T_e/T_i (PBI-C-0004)

fit, region	certified θ^*	witness at θ_{lo}
NS, full box	[0.4223, 0.4308]	$T_i = 243.5$ keV, $x = 0.0922$, $R \geq 1.0077$ at $\theta = 0.4223$
NS, $x = 1/6$	[0.3753, 0.3828]	243.5 keV, $R \geq 1.0082$ at $\theta = 0.3753$
Wang, full box	[0.4381, 0.4470]	243.5 keV, $x = 0.092$, $R \geq 1.0082$ at θ_{lo}
Wang, $x = 1/6$	[0.396, 0.402]	**261.65** keV, $x = 1/6$, $R \geq 1.00037$ at $\theta = 0.396$ (thin margin; audit C8)

At θ_{hi} the certified $\max R$ is below 1 (for example ≤ 0.99999999 for Wang over the box and ≤ 0.99995 for Wang at $x = 1/6$, $\theta = 0.402$). The Wang value at $x = 1/6$ comes from the explicit θ scan `e1_wang_mix.py` (`logs/certify_wang_mix.json`): at $\theta = 0.398$ and 0.400 neither a witness nor a certificate exists, at 0.402 the certificate does. The auditor's float checks put the thresholds at 0.4287, 0.3811, 0.4451, 0.3988, each inside the certified interval; the auditor's witness for Wang at $x = 1/6$ gives $R = 1.0015$ (float) against the producer's certified lower bound 1.00037. Interval arithmetic certifies the NS upper side and the NS witnesses on the auditor's side; the Wang side is float there.

Scope, with the audit's qualification (C4, C5). Under these two fits and Putvinski's bremsstrahlung, a window needs T_e/T_i below about 0.42–0.45 (box) or 0.375–0.402 ($x = 1/6$). θ^* **is not a property of** p-^{11}B**; it is a property of the chosen reactivity and bremsstrahlung fits.** Auditor sensitivities (float, Tier [X]): uniform reactivity $\times 1.1/1.2/1.32$ gives $\theta_c = 0.477/0.524/0.579$ (the factor 1.32 is an illustrative product of "+20% cross section, +10% kinetic", not a pinned value); Rider's bremsstrahlung moves θ_c by about +1% (to 0.434 NS, 0.452 Wang) and raises $\max R$ at $\theta = 1$ to 0.438 and 0.447; $E = 8.682$ MeV gives 0.4277; removing Wang's 4.7 MeV resonance moves 0.4451 to 0.4450. The maximum over the box for $\theta < 1$ lies at $T_i \approx 235$–259 keV, $x \approx 0.093$.

5.3 Remark on Fig. 28 of Wurzel–Hsu

The caption of Fig. 28 of [1] reads "P_B always exceeds P_c when $T_e \gtrsim T_i/3$", with the parameterized Nevins–Swain reactivity and the $x = 1/6$ mix. In the same model the certified threshold is $[0.3753, 0.3828]$, not $1/3$; so for T_e/T_i between $1/3$ and about 0.375 there are temperatures, near $T_i \approx 243$–248 keV, with $P_c > P_B$. By float check at $T_e = T_i/3$, $x = 1/6$: $\max P_c/P_B = 1.113$ at 248.3 keV (Nevins–Swain) and 1.161 at 268.5 keV (Wang) (`logs/third_check.txt`; the auditor finds 1.1132 and 1.1606; Tier [X]). This **sharpens a figure reading and does not contradict their conclusions**: the caption's "$\gtrsim$" makes it a rough reading of a plotted curve, and their statement that p-^{11}B is not ignitable at $T_e = T_i$ is what we also find.

6 Controls

control	expectation	outcome
P1 D-T	gate PASS	PASS (G1, G2, G2b, G3, G4 true), for the authors' code model
P2 float	inside the certified hull	inside: $T_{\rm ign} = 4.29959$, TP min 2.94518×10^{21}, LP min 1.55284×10^{20} (separate scipy code; text C_B/J, *code* $f_c = 0.2$, relabelled per C1)
P3	W–H Tables II/III "–" for p-^{11}B (NS, $T_{\rm e} = T_{\rm i}$, $x = 1/6$)	certified max $R \leq 0.39222$
P4 float vs p-^{11}B	float inside the certified intervals	inside, all six quantities (Section 5)
N1	reactivity $\times 0.5$: gate FAIL	FAIL: $T_{\rm ign} = 5.3724$ keV, TP min 6.285×10^{21}, LP min 3.230×10^{20}
N2	bremsstrahlung removed: conclusion flips, gate FAIL	FAIL: no ignition temperature (every T has $S_c > S_B$), TP min 2.758×10^{21}; the p-^{11}B ratio is infinite
N3 Lean	each planted statement fails to elaborate	rc 1, errors at three declarations (Section 7)
S positivity	needed for the T-interval bound	certified for both Bosch–Hale pieces and the three Wang segments

What the D-T gate does and does not show (audit C6). The D-T gate G3 does not discriminate for the p-^{11}B conclusion: it still passes under N1, because halving the reactivity only strengthens an exclusion. The p-^{11}B pipeline's ability to *find* a window is shown instead by the θ^* witnesses ($R > 1$ at $\theta_{\rm lo}$), by the check at $\theta = 1/3$ ($R \approx 1.11$) and by N2. The auditor read the recorded check vectors of N1 and N2 in `logs/certify.json`; the gate code itself was not re-run by the auditor (`PBI-C-0005`).

7 Lean (Tier [A] pending a statement lock)

Module `formal_cert/Programmes/PbIgnition/Core.lean` (copy in `data/lean/`), `import Mathlib.Tactic` and `Mathlib.Analysis.SpecialFunctions.Pow.Real` (never `import Mathlib`), Lean 4.34.1, Mathlib `d13f23b7`, through `scripts/certified_numerics/lean_pinned.py`, rc 0, no `sorry`, no warnings. Theorems: `optimal_mix` ($20x(1-x) \leq (1+4x)^2$ for all real x; with $x = n_B/n_i$ this is $n_p n_B/n_e^2 \leq 1/20$), `optimal_mix_eq` (equality iff $x = 1/6$; W–H footnote 138), `rpow_25_le_sq` ($t^{2.5} \leq t^2$ on $[0,1]$) and `ee_term_deriv_pos` ($0 < 1 + 2.2t + 3t^2 - 4.375t^{2.5}$ on $[0,1]$; $4.375 = 3.5 \times 1.25$ is the exact coefficient of the derivative of $t(1 + 1.1t + t^2 - 1.25t^{2.5})$). `#print axioms` for all four: `propext`, `Classical.choice`, `Quot.sound` (`logs/lean_core.json`); the binding `anse/formal/lean_runner.py verify_file` gate, with `lean_cmd` pointed at the pinned toolchain, gives the same (`logs/lean_runner_core.json`; the main `formal/` build lacks `Mathlib/Tactic.olean`, so its default command fails before any proof is checked). Escape grep (`sorry`, `admit`, `native_decide`, `axiom`, `set_option`, `#exit`, `skipKernelTC`, `addDecl`, `implemented_by`, `unsafe`, `opaque`, `macro`, `decide`, ...): no code hit (the word "elaborate" in a docstring of `Controls.lean` only).

Independent verification (audit C3). The auditor recompiled a byte-identical copy with `lean_pinned.py compile` (rc 0, 5.5 s), built its own olean and ran `leanchecker PbIgnition.Core` (rc 0, 40.2 s); a nonexistent module gives rc 1 (negative control), so the rc 0 is a real replay. **The leanchecker transcript is in the auditor's scratch and is not shipped;** it is reported in `AUDIT.md` §5 only. The binding `lean_runner` gate was run by the producer, not re-run by the auditor. **No statement-lock entry exists**; so the two facts are *kernel-checked, independently recompiled and leanchecker-replayed, and become Tier [A] once a statement-lock entry exists.* In the ledger they are filed as Tier [A] with an `ai_audit` field and the gate field `audit` left null, exactly as in the sibling papers, so the gate flags them.

Controls. `Controls.lean` (tampered optimum 1/21 with the same proof script; tampered coefficient 7 instead of 4.375; the derivative bound claimed on $[0,2]$) is rejected, rc 1, with errors at lines 23, 28, 32, 33

(logs/lean_controls.json). The auditor checked that each planted statement is actually false (at $x = 1/6$: $21 \cdot 5/36 = 2.917 > 25/9$; at $t = 1$: $1 + 2.2 + 3 - 7 = -0.8$; at $t = 2$: about -7.35), so no proof exists, not merely that the script fails (PBI-A-0002).

What the Lean says, and only this. Two elementary real inequalities used by the model: the optimum of the proton–boron mix, and the sign of the derivative of the electron–electron bremsstrahlung term on $[0, 1]$. **No interval enclosure, no reactivity, no ignition statement is in Lean**; neither is the monotonicity step. The enclosures of Sections 4–5 are Tier [C].

- Core.lean: 34c301362049496f... (full sha256 in ledger/evidence/, data/logs/sha256.json)
- Controls.lean: d95715aa7324ca55...

8 Audit corrections applied

The independent audit (AUDIT.md §8, a different AI agent instance, not a person) required nine corrections to the producer's report. Where the producer's RESULT.md and result.json (copied unchanged into data/) differ, **this paper supersedes them**.

C1. The D-T "pure text" variants are relabelled "text C_B/J, code $f_c = 0.2$" (they use the code's f_c, FC_CODE in e1_certify.py); the true pure-text row ($f_c = 3.5/17.6$) is added: $T_{\rm ign} \in [4.3071225, 4.3071227]$ keV and the triple-product minimum in $[2.95 \times 10^{21}, 2.96846 \times 10^{21}]$, printing 3.0×10^{21} (auditor, interval for $T_{\rm ign}$ and for the lower bound; Tier [X] here, not certified by the producer pipeline). Control P2 is relabelled likewise.

C2. Gate status: "passes for the model the authors' code ran (the gate model was chosen after a float run, Deviation 6)"; the most literal text reading prints 3.0×10^{21}.

C3. Lean: "kernel-checked, independently recompiled and leanchecker-replayed by the auditor; Tier [A] once a statement-lock entry exists". Only the statement lock is missing. (result.json → tiers.lean_lemmas still reads "A pending independent verification"; this paper supersedes it.)

C4. The robustness sentence now reads "under these two fits and Putvinski's bremsstrahlung, a window needs $T_{\rm e}/T_{\rm i}$ below about 0.42–0.45", with the reactivity scaling (×1.2 gives 0.52) and Rider's bremsstrahlung (+1%).

C5. The no-window statement carries its model qualifiers (no synchrotron, no transport, Maxwellian, no $T_{\rm e}$-maintaining mechanism; box $T_{\rm i} \in [50, 500]$ keV, $x \in [0.05, 0.5]$; interior maximum; widening the box changes nothing, float).

C6. The D-T gate does not discriminate for the p-^{11}B conclusion (Section 6).

C7. **Third-party material is not in the repository or in this bundle**: the Wurzel–Hsu full text and code at 4ae563bf, and the paywall and challenge pages, were moved out of the repository on 2026-10-28 and can be re-fetched from the sources in data/fits.json. result.json, logs/sha256.json and RESULT.md §10 still list data/papers/* and data/downloads/* entries and hashes: **those files are no longer in the repository** (the hashes identify them; the auditor verified that three match the moved copies). code/e1_hashes.py lines 32–34 still point at the removed data/downloads path and would fail if re-run; the script is shipped unmodified so that its recorded hash stays valid.

C8. The Wang $x = 1/6$ witness is recorded: $T_{\rm i} = 261.6496$ keV, $x = 1/6$, certified $R \geq 1.00037$ at $\theta = 0.396$ (logs/certify_wang_mix.json); the margin is thin, the auditor's float value is 1.0015.

C9. Neither producer nor auditor obtained Bosch–Hale 1992 or Nevins–Swain 2000; the coefficients remain secondary (author code only) (Section 2).

9 Claim ledger

ledger/build_ledger.py files 14 claims: 2 at Tier [A] (PBI-A-0001, -0002), 3 at [L] (PBI-L-0001..0003; -0003 is the unverified coefficients), 7 at [C] (PBI-C-0001..0007, the last being the headline) and 2 at [X]

(PBI-X-0001: float checks and the auditor's float and pure-text values; PBI-X-0002: what was not run). Each carries a content-addressed evidence blob in ledger/evidence/; each Tier [A] claim also carries an ai_audit field. The gate field audit is left null: the audit was done by an AI agent, not a person, so the Elenchus gate keeps LEDGER_UNAUDITED_TIER_A open. The gate (tools/ledger.py ledger.json --evidence-dir evidence; verbatim in ledger/gate_output.txt) exits with code 1: 0 blocking findings and 2 flags, one per Tier [A] row. That is the honest state.

10 Limitations

- **Model, fits and box.** Every statement is about an idealised 0-D power balance with the stated fits in $T_\mathrm{i} \in [50, 500]$ keV, $x \in [0.05, 0.5]$. Ideal ignition only: no conduction and no confinement time; it is not reactor breakeven, which is a different and weaker condition (Ochs–Fisch [3]; their breakeven and ignition-time figures come from a four-species optimisation not reproduced here).
- **Not in the model:** synchrotron radiation (it would only make every window smaller); transport and conduction losses; any mechanism maintaining $T_\mathrm{e} < T_\mathrm{i}$, that is the ion–electron collisional coupling a self-consistent T_e would need; α heating of electrons; ash and impurities; non-Maxwellian or kinetic enhancement (bounded only through m^*); the uncertainties of the fits (Wang quote ±3–10% on the 0.6 MeV resonance; the Nevins–Swain errors are not propagated).
- **The θ^* results do not say that $T_\mathrm{e} < T_\mathrm{i}$ is achievable**, and are a property of the chosen fits (Section 5).
- **Coefficients not verified against primary papers** (Bosch–Hale 1992, Nevins–Swain 2000; Section 2); the Putvinski bremsstrahlung equation was seen only through OCR; Tentori–Belloni and Sikora–Weller were not used as fits. "Updated fit" means Wang 2026 only.
- **Tier [C], not Tier [A].** The enclosures are Arb ball arithmetic not replayed in Lean (the preregistration asked for Tier [A]); the monotonicity step is not formalised. No human audit; AI audit only. The auditor's Wang-side numbers are float, not rigorous; its D-T minimisers of the gate model are float; the producer's Arb integrals were not re-run by the auditor.
- **Lean:** two small facts only; recompilation and replay by the auditor; no statement lock; the auditor's leanchecker transcript is not shipped.
- **Deviations from the preregistration** (preregistration.json was not edited): (1) sources: coefficients from the Wurzel–Hsu code (the listed fallback), so the kill criterion's wording "pinned from primary sources" is not met; (2) the updated fit is Wang 2026, not Sikora–Weller or Putvinski; (3) Tier [C] instead of [A] for the inequalities; (4) θ is certified on $[\theta_\mathrm{hi}, 1]$ only, not for $\theta > 1$; (5) E1c is answered twice, as an exclusion under both fits and as the multiplier m^*; (6) the gate was written after a first float run and tests the code path that produced Table III; (7) certification-method changes between runs, with run logs kept, and the early-exit bug and unfixed stopping order described in Section 3; (8) not done: the CVODE transient (optional) and a synchrotron or kinetic sensitivity.
- **Repository guards.** antigravity_guard.py (38 findings, none in E1 files) and test_rigor_guard.py exit 1 on pre-existing files elsewhere; pytest tests/ was not run (E1 adds no tests) (data/logs/guard_*.txt).
- **Novelty.** No priority claim; to our knowledge nobody has published these enclosures, but this was not searched systematically.
- Public data only; no inertial-confinement, burn or yield content. Not a statement about any device, plant, climate or economics.

11 Reproducibility

Repository AutoevolveAI, branch worktree-openai-math-discovery; HEAD when this paper was written: bdd3e02b174fe4a2c1751431e0d5b573a86d3dc9. Copies of the inputs are in data/ (RESULT.md, result.json, AUDIT.md, fits.json, code/, logs/, lean/); every file is below 5 MB. The Wurzel–Hsu code and papers are not included; to re-run, fetch them as described in fits.json and restore data/downloads/ (the Arb model

reads only `fits.json`). The command lines below are summarised from the run note and code headers; they were not re-executed in the paper bundle.

```
taskset -c 6,7 python code/e1_certify.py           # gates, D-T variants, pB stages (python-flint 0.6.0, Arb 128 bit)
python code/e1_wang_mix.py                          # Wang theta* at x = 1/6 (explicit scan)
python code/e1_float_indep.py ; python code/e1_ns_consistency.py ; python code/e1_third_check.py   # Tier X
python -I scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/PbIgnition/Core.lean
python -I papers/p4_pb_ignition/ledger/build_ledger.py ; python -I papers/p4_pb_ignition/ledger/run_gate.py
```

References

[1] S. E. Wurzel, S. C. Hsu, *Progress toward fusion energy breakeven and gain as measured against the Lawson criterion*, arXiv:2105.10954 (v4). The model, Tables II/III, Fig. 28 and the public code at github.com/swurzel/lawson-criterion-paper, commit `4ae563bf` (MIT licence) ([L]).

[2] Wang, Li, Wu, Cui, *Revisiting p-^{11}B Fusion: Updated Cross-sections, Reactivity, and Energy Balance*, arXiv:2601.00241 (v1, 2026). S-factor, Table I, eqs. (1)–(6), Fig. 8 text ([L]).

[3] I. E. Ochs, N. J. Fisch, arXiv:2310.18508 (v1, 2023), eq. (14) (bremsstrahlung with the x^2 term); breakeven and ignition-time figures out of scope ([L]).

[4] H.-S. Bosch, G. M. Hale, *Improved formulas for fusion cross-sections and thermal reactivities*, Nucl. Fusion 32, 611 (1992). **Not obtained by producer or auditor**; coefficients via the Wurzel–Hsu code.

[5] W. M. Nevins, R. Swain, *The thermonuclear fusion rate coefficient for p-^{11}B reactions*, Nucl. Fusion 40, 865 (2000). **Abstract only; not obtained**; coefficients via the Wurzel–Hsu code.

[6] M. H. Sikora, H. R. Weller, J. Fusion Energy 35, 538 (2016); read, not used as a fit.

[7] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019); cited for the library only.